0054. Spiral Matrix

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Problem

- Return all elements of an m x n matrix in clockwise spiral order.

Intuition

- Keep four boundaries (top, bottom, left, right). Walk the top row, right column, bottom row and left column, shrinking a boundary after each side.

Brute Force

- Simulate with direction changes and a visited matrix: O(mn) time and extra space.

Optimal Approach

- While top <= bottom and left <= right:
- Top row left to right; top++. Right column top to bottom; right--.
- If top <= bottom: bottom row right to left; bottom--. If left <= right: left column bottom to top; left++.

Worked Example

```
1  2  3  4
5  6  7  8      -> 1 2 3 4 8 12 11 10 9 5 6 7
9 10 11 12
```

Java Solution

```java
public class SpiralMatrix {
    public List<Integer> spiralOrder(int[][] m) {
        List<Integer> res = new ArrayList<>();
        int top = 0, bottom = m.length - 1, left = 0, right = m[0].length - 1;
        while (top <= bottom && left <= right) {
            for (int c = left; c <= right; c++) res.add(m[top][c]);
            top++;
            for (int r = top; r <= bottom; r++) res.add(m[r][right]);
            right--;
            if (top <= bottom) {
                for (int c = right; c >= left; c--) res.add(m[bottom][c]);
                bottom--;
            }
            if (left <= right) {
                for (int r = bottom; r >= top; r--) res.add(m[r][left]);
                left++;
            }
        }
        return res;
    }
}
```

Complexity

- Time: O(m n). Space: O(1) besides the output.

Pitfalls

- The two inner checks prevent visiting a single remaining row or column twice.